

LESSON 325 (2.0 GROUND)

Lesson Objective

During this ground lesson, the student's knowledge of weather, aircraft systems, and flight instruments will be reviewed. This lesson will also introduce the student to IFR cross-country planning.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

References

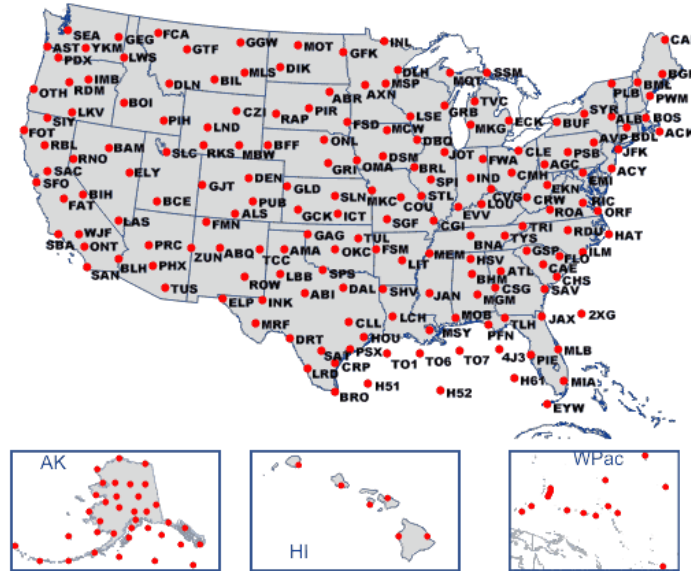
- Advisory Circulars. ([Search Tool](#))
 - Aviation Weather. ([AC 00-6B](#))
 - Aviation Weather Services. ([AC 00-45H](#))
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 13 - Aviation Weather Services.
 - Chapter 15 - Airspace.
 - Chapter 16 - Navigation.
- Instrument Procedures Handbook ([FAA-H-8083-16B](#))
- Aeronautical Chart Users' Guide ([FAA](#))
- Instrument Flying Handbook ([FAA-H-8083-15B](#))
- Aviation Weather Handbook ([FAA-H-8083-28](#))

Lesson Content

Review: Obtaining Weather Information

- Weather briefings:
 - Types of weather briefings:
 - Outlook.
 - Standard.
 - Abbreviated.
 - In-flight.
- METAR - Aviation routine weather report:
 - Observation of current surface weather.
 - Issued hourly unless significant changes have occurred (like wind direction changing by 45 degrees or more in less than 15 minutes, more in AC).
 - A special METAR (SPECI) can be issued between hourly reports.
 - Know how to decode them. Use the METAR decoding help sheet if needed.
- TAF - Terminal aerodrome forecast:
 - Established for a 5SM radius around an airport.
 - Valid for 24 up to 30 hours.

- Updated 4 times a day.
- Know how to decode them. Use the METAR decoding help sheet if needed.
- **PIREPs - Pilot weather reports.**
 - Weather report from a flight deck. It consists of message type, location, time, altitude/flight level, type of aircraft, and at least one other element to describe the report phenomena.
 - 2 types: routine (UA) and urgent (UUA):
 - Urgent includes:
 - Tornadoes.
 - Severe or extreme turbulence.
 - Hail.
 - Low-level wind shear within 2,000ft.
 - Required conditions to issue a PIREP:
 - Ceilings at or below 5,000 feet.
 - Visibility reported on the surface or aloft is 5 miles or less.
 - Thunderstorms and related phenomena.
 - Turbulence of moderate degree or greater.
 - Icing of light degree or greater.
 - Wind shear.
 - Braking action reports less than good.
 - Volcanic eruption, ash clouds, and/or detection of sulfur gasses in the cabin.
- **GFA Tool - Graphical forecasts for aviation.**
 - Show the student where to find the GFA tool.
 - Go over the different items that the GFA tool can display:
 - Visibility, ceilings, cloud coverage, METARs, TAFs, precipitation, winds, thunderstorms, low-level wind shear, turbulence, and icing.
- **Winds and temperatures aloft chart (Fb)**
 - A summary of winds aloft forecast for the route/area as interpolated from forecast data for the local and/or the adjacent reporting locations for level through 12,000 feet.
 - Winds and temps at specific locations.
 - On a global scale, Issued 4 times a day (Every 6hrs) with forecasts out to 6, 12 & 24hrs.
 - The wind is in true knots.
 - The temperature is in Celsius.
 - If the wind is forecasted to be 100-199 knots, add 50 to the wind direction and subtract 100 from the speed.



- 731960
 - Wind is heading 230 at 119 knots.
 - The Temperature is -60 C.
- If the wind is at least 200 knots, the wind will be coded as 99 knots.
 - 7799.
 - The wind is 270 and greater than 199 knots.
- Calm or less than 5 knots is 9900.
- NOTE:
 - No winds are forecasted within 1,500 feet AGL of station elevation.
 - No temperatures are forecasted within 2,500 feet AGL of station elevation.
- Quiz the student on determining the atmosphere's stability using this chart!
- Foreflight Winds/Aloft
 - Receives winds aloft forecast data from the Foreflight Global Wind Engine, providing winds and temperature data at any point on the planet up to FL540.
 - The Foreflight Global Wind Engine blends in observed data from the National Weather Service to accurately forecast turbulence, icing, winds & temperature across the world.
- Radar Imagery:
 - National radar:
 - Shows the intensity of precipitation.
 - Provides a single view of the entire country.
 - Updated every 10 minutes.
 - Loops the latest 2 hours of images.
 - Single site radar:
 - Composed of 140 radar sites across the U.S.
 - Each site shows base reflectivity, composite reflectivity, and total precipitation.
 - Updated every 10 minutes.

- Loops the latest 2 hours of images.
- ADS-B Radar imagery
 - ADS-B radar imagery has a lower resolution compared to internet radar due to bandwidth limitations. Similarly, regional radar imagery (within 250nm) has a higher resolution than the CONUS imagery.
- Significant weather prognostic chart:
 - Low-level significant weather-surface to FL240.
 - High level - FL250 to FL600.
 - 2 forms:
 - 12 and 24-hour forecast chart:
 - Issued 4 times a day.
 - The valid time is printed on the chart.
 - Shows forecasted weather to include non-convective turbulence, freezing levels, and IFR or MVFR weather.
 - Cover the difference between IFR, MVFR, and VFR.
 - IFR: less than 1000ft ceilings and less than 3SM visibility.
 - MVFR: 1000ft - 3000ft ceiling and 3-5SM visibility.
 - VFR: more than 3000ft ceiling and more than 5SM visibility.
 - 36 and 48-hour surface forecast chart:
 - Extension of the above chart.
 - Surface weather forecasts.
 - Issued 2 times a day.
 - Pressure patterns, fronts, and precipitation.
- Surface analysis chart:
 - Surface analysis charts – Chart depicts the distribution of several items, including sea level pressure, the position of the highs, lows, ridges, and troughs, the location and type of fronts, wind direction and speed, temperature, and dew point.
 - Analysis of current weather.
 - Computerized.
 - Transmitted every 3 hours.
 - Valid time is shown on the chart (observed data).
- AIRMETs, SIGMETs, and convective SIGMETs:
 - Airmet vs G-AIRMET
 - G-AIRMETs replaced the legacy textual airmets back in 2010.
 - G-AIRMET is a snapshot or a graphical depiction of a hazard at that given time using lat/long coordinates, whereas an AIRMET is a textual description using VORs to describe the affected area.
 - Before G-AIRMETs existed, pilots had to manually draw the affected area on a sectional chart after receiving a weather briefing.
 - G-AIRMETs are issued every 3 hours, whereas traditional AIRMETs are issued every

6 hours. AIRMETs are generated by the software using the initial, 3hr & 6hr forecast.

- AIRMET (WA) – Advises of light to moderate weather that may be hazardous to aircraft (primarily light).
 - Issued every 6 hours.
 - Valid for 6 hours.
 - Sierra:
 - IFR conditions (ceilings less than 1000 ft, visibility less than 3sm).
 - Mountain obscurations.
 - Tango:
 - Light to moderate Turbulence.
 - Surface winds 30 knots or greater.
 - Low-level wind shear.
 - Zulu:
 - Icing.
 - Freezing levels.
 - An AIRMET will be issued if any of these conditions are expected to happen over an area of at least 3,000 square miles.
- SIGMET (WS) – Advises of severe weather conditions that may be hazardous to all aircraft.
 - Unscheduled (comes out when needed).
 - Valid for 4 hours (6 for hurricanes).
 - Severe icing not associated with thunderstorms.
 - Severe or extreme turbulence or CAT not associated with thunderstorms.
 - Dust/sandstorms.
 - Volcanic ash.
 - Issued with an alphabetic identifier (no sierra or tango) → different zones throughout the USA have designated identifiers.
 - Issued when these weather conditions are forecasted to happen over an area of at least 3000 square miles or an area judged to have a significant impact on aircraft safety.
- Convective SIGMET (WST) – Hazardous convective weather affecting every flight.
 - Only issued for the contiguous United States.
 - Issued at 55 minutes past the hour.
 - Valid for 2 hours.
 - Severe thunderstorms with surface winds greater than 50 knots
 - Hail at the surface at least 3/4 inch in diameter
 - Affects 40% or more of a 3000 sq miles area.
 - Line of thunderstorms at least 60 miles long with thunderstorms affecting at least 40 percent of its length.
 - Tornadoes
 - Numbered sequentially depending on the zone (West, Central, East).

- If there is no convective weather, the chart will say “NONE”.
- Center Weather Advisory:
 - A CWA is an aviation weather warning for conditions meeting or approaching national in-flight advisory (AIRMET, SIGMET, or Convective SIGMET) criteria.
 - CWAs are valid for up to 2 hours and may include forecasts of conditions expected to begin within 2 hours of issuance.
 - If conditions are expected to persist after the advisory’s valid period, a statement to that effect is included in the last line of the text. Additional CWAs will subsequently be issued as appropriate
 - CWA are issued by the NWS CWSU (Center Weather Service Unit). The 2 centers in Florida are Jacksonville & Miami.
- Convective outlook:
 - Consists of both a narrative and a graphic showing severe thunderstorm threats across the US.
 - Convective outlooks are split into 4 different charts:
 - Day 1: risk of severe weather for 24 hours. Issued 5 times a day.
 - Day 2: risk of severe weather for the next 24 hours. Issued 2 times a day.
 - Day 3: risk of severe weather for the next 24 hours. Issued daily.
 - Day 4-8: risk of severe weather for day 4 through day 8. Issued daily.
 - <https://www.weather.gov/jetstream/ahead>
- AWOS, ASOS, and ATIS:
 - Check the chart supplement to see what type of weather system is in use.
 - AWOS:
 - Automated Weather Observing System.
 - Operated and controlled by the FAA.
 - Automated and reports the weather every minute.
 - Types of AWOS:
 - AWOS-A: Altimeter setting.
 - AWOS-AV: Altimeter and visibility.
 - AWOS-I: Altimeter, wind, temperature, dew point, and density altitude.
 - AWOS-II: AWOS-I plus visibility.
 - AWOS-III: AWOS-II plus ceiling.
 - AWOS-III P: AWOS-III plus precipitation identification.
 - AWOS-III T: AWOS-III plus thunderstorm/lightning identification.
 - AWOS-III PT: AWOS-III plus precipitation and thunderstorm/lightning identification.
 - AWOS-IV: AWOS-III plus precipitation, type and accumulation, freezing rain, thunderstorm, and runway surface sensors.

- ASOS:
 - Automated Surface Observing System.
 - Operated and controlled by the National Weather Service (NWS), Department of Defense (DoD), and sometimes by the FAA.
 - Used for the national weather system, not just for aviation purposes.
 - Reported information:
 - Wind speed and direction.
 - Visibility.
 - Sky condition.
 - Ceiling height and precipitation.
 - Barometric pressure.
 - Density altitude.
- ATIS:
 - Automated Terminal Information Service.
 - Towered airports.
 - Requires a human to monitor and broadcast the ATIS.
 - ATIS information is obtained from the ASOS system.
 - Contains more than just weather (NOTAMs, remarks, etc.)
 - Published hourly at 55 minutes past the hour.
 - If the weather is rapidly changing, the tower can give an updated ATIS (like a SPECI METAR).
 - Reported information:
 - Phonetic alphabet letter.
 - Time of the report.
 - Wind speed and direction.
 - Visibility.
 - Cloud cover.
 - Temperature.
 - Dew point.
 - Barometric pressure.
 - NOTAMs, remarks, other information.

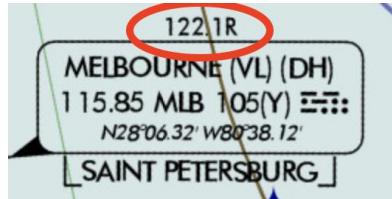
Introduce: IFR Cross-Country Completing and Filing

- 91.103 Preflight action - IFR flight or away from the vicinity of an airport:
 - NOTAMs.
 - Weather.
 - Known delays.
 - Runway lengths.
 - Alternates if needed.
 - Fuel requirements.
 - Takeoff/landing data.

Flight Plan: Completing and Filing

Demonstrate the process of filing an IFR flight plan using one of these methods.

- Filing a flight plan:
 - Flight Service Station (FSS)
 - Phone: 1-800-WX-BRIEF
 - Radio:
 - Frequencies are shown in chart supplement or on the LE chart.



- RCO/GCO - check chart supplement.
 - Online: 1800wxbrief.com
- EFB
 - Foreflight: Flights tab
- Structure:
 - Flight rules: I (IFR)
 - Equipment (FITA fleet): SG / LB1
 - S: Standard navigation/communication equipment (VOR, VHF, ILS)
 - G: GNSS navigation
 - L: Mode S Transponder with Altitude, Enhanced Surveillance, Extended Squitter
 - B1: 1090 MHz (ADS-B) Out
 - Route: input route of flight only. Do not include departure/destination airports.
 - Alternate: input alternate airport. Requirements will be reviewed below.
 - Option to include 2nd alternate.
 - Emergency radio: VHF & ELT.
 - Jackets: Fluorescein (if bringing life jackets).
 - Remarks: "NO SID/STAR" if unwilling/unable to accept these procedures.

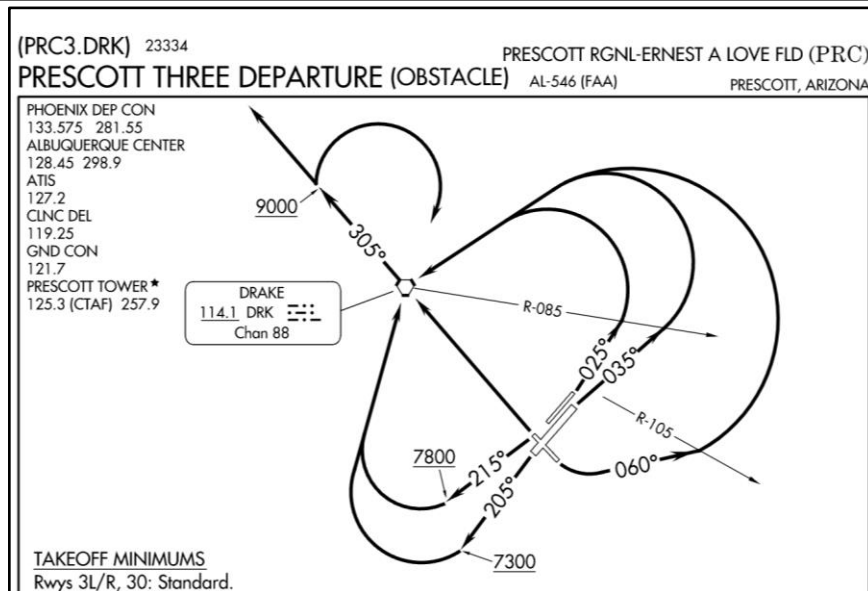
Departure Procedures

This section should be a review from lesson 318. Consider quizzing the student to evaluate their knowledge.

- Departure procedures:
 - Provide a transition from the airport to the enroute structure while ensuring obstruction clearance.
 - Pilots operating under Part 91 are encouraged to file and fly a departure procedure when one is available.
 - Criteria required to provide obstruction clearance on a departure procedure (AIM 5-2-9):
 1. Climb at 200ft/nm until told to level off or upon reaching the minimum IFR altitude.
 2. Cross the departure end of the runway (DER) after reaching 35' AGL.
 3. Make the initial turn only after reaching 400' AGL.

- Types of Departure Procedures:
 - Obstacle Departure Procedure (ODP)
 - Printed either textually or graphically.
 - Textual ODPs will be listed under DEPARTURE PROCEDURE in the IFR Takeoff Minimums, Departure Procedures, and Diverse Vector Area (Radar Vectors) section of the TPP.
 - Graphical ODPs will include (OBSTACLE) in the title and will be found in the Terminal Charts section of the TPP.
 - Provides obstruction clearance from the terminal area to the enroute structure.
 - Not mandatory to follow but recommended (e.g. night, IMC, terrain).
 - Does not require an ATC clearance unless a SID or radar vectors has been assigned by ATC.

ST PETERSBURG, FL
ALBERT WHITTED (SPG)
TAKEOFF MINIMUMS AND (OBSTACLE) DEPARTURE PROCEDURES
AMDT 4 20APR23 (23110) (FAA)
TAKEOFF MINIMUMS:
Rwy 18, 300-1.
Rwy 36, 300-1 or std. w/min. climb of 845' per NM to 300.
DEPARTURE PROCEDURE:
Rwy 7, climb on heading 067° to 500 before turning left.
Rwy 18, climb on heading 185° to 500 before turning right.
Rwy 25, climb on heading 247° to 700 before turning right.
Rwy 36, climb on heading 005° to 700 before turning left.

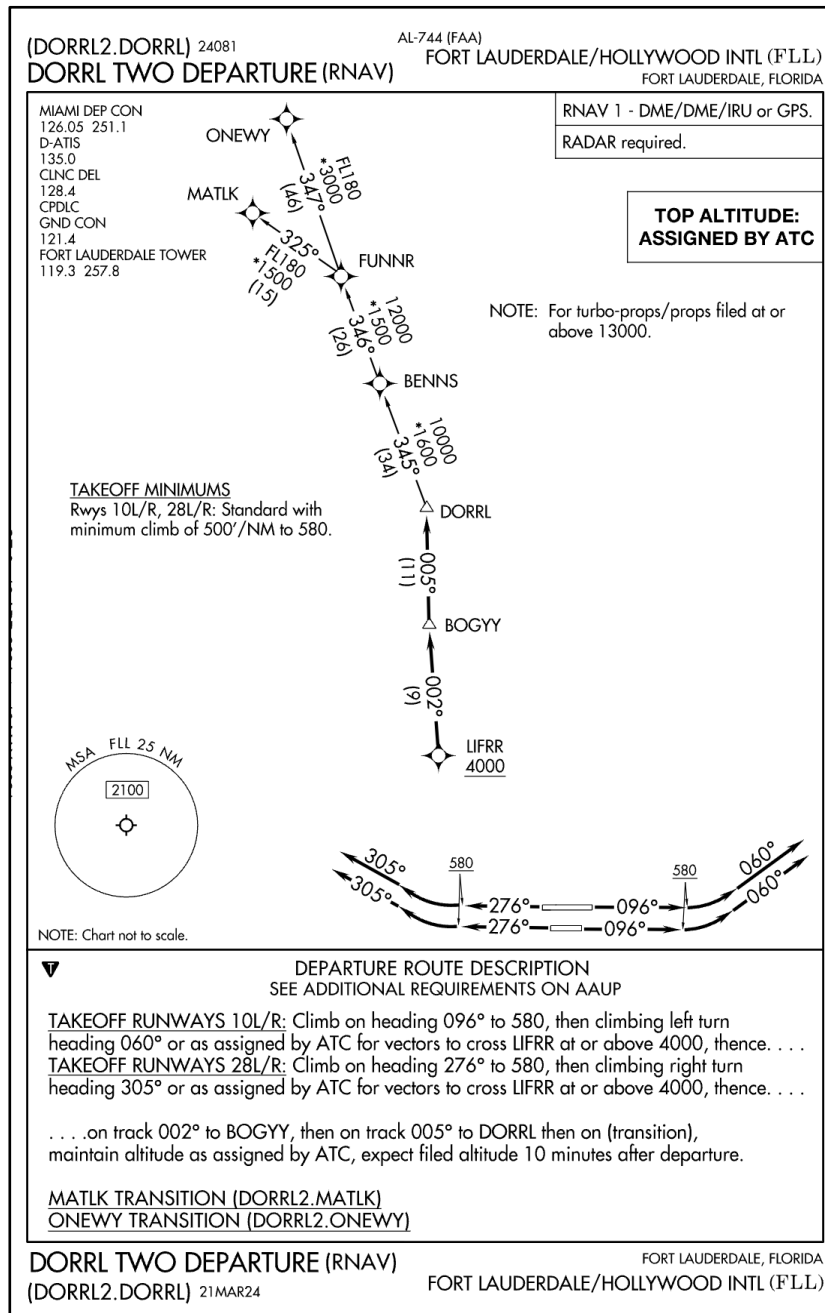


- Standard Instrument Departure (SID)
 - Graphical procedures printed for pilots and controllers.
 - Provides obstruction clearance from the terminal area to the enroute

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structure.

- Designed to reduce pilot/controller workload.
- Requires an ATC clearance.
- Can be found in the Terminal Charts section of the TPP, just after the approach procedures for a particular airport.
- Go through the symbology of a SID.
- Demonstrate how to find and brief a SID.



■ Diverse Vector Areas

- A Diverse Vector Area (DVA) is an area in which ATC may provide radar vectors until the aircraft reaches a safe altitude.

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- Pilots should be aware that Air Traffic facilities may utilize a climb gradient greater than the standard 200 ft/NM within a DVA.
 - See example below for KDVT; upon receiving radar vectors after departure, pilots have to follow the climb gradient (CG) listed in the DVA.



**TAKEOFF MINIMUMS, (OBSTACLE) DEPARTURE PROCEDURES, AND
DIVERSE VECTOR AREA (RADAR VECTORS)**



24081

PHOENIX, AZ (CON'T)
PHOENIX DEER VALLEY (DVT) (CON'T)

Rwy 25L (CON'T), tree, building, antenna, pole beginning 542' from DER, 502' right of centerline, up to 39' AGL/1475' MSL.
 Tree, building, pole beginning 630' from DER, 505' left of centerline, up to 1467' MSL.
 Pole 1070' from DER, 375' left of centerline, 1468' MSL.
 Poles beginning 1077' from DER, 374' left of centerline, up to 39' AGL/1470' MSL.
 Pole 1126' from DER, 498' right of centerline, 44' AGL/1478' MSL.
 Poles beginning 1183' from DER, 495' left of centerline, up to 41' AGL/1471' MSL.
 Signs beginning 1234' from DER, 356' left of centerline, up to 48' AGL/1478' MSL.
Rwy 25R, lighting 30' from DER, 114' left of centerline, 3' AGL/1456' MSL.
 Lighting, general utility beginning 30' from DER, 112' right of centerline, up to 2' AGL/1456' MSL.
 General utility 31' from DER, 112' left of centerline, 4' AGL/1457' MSL.
 Tree 1139' from DER, 798' right of centerline, 1490' MSL.
 Terrain 1.3 NM from DER, 2396' right of centerline, 1686' MSL.
 Vegetation, tree beginning 1.3 NM from DER, 2270' right of centerline, up to 1697' MSL.
 Pole, traverse way beginning 1.3 NM from DER, 1847' right of centerline, up to 30' AGL/1713' MSL.
 Antenna 1.3 NM from DER, 2640' right of centerline, 57' AGL/1732' MSL.
 Tower, tree, pole, vegetation beginning 1.3 NM from DER, 2073' right of centerline, up to 66' AGL/1741' MSL.
 Antenna, tower, traverse way, tree beginning 1.4 NM from DER, 2749' right of centerline, up to 91' AGL/1769' MSL.
 Tree 1.6 NM from DER, 3023' right of centerline, 25' AGL/1700' MSL.

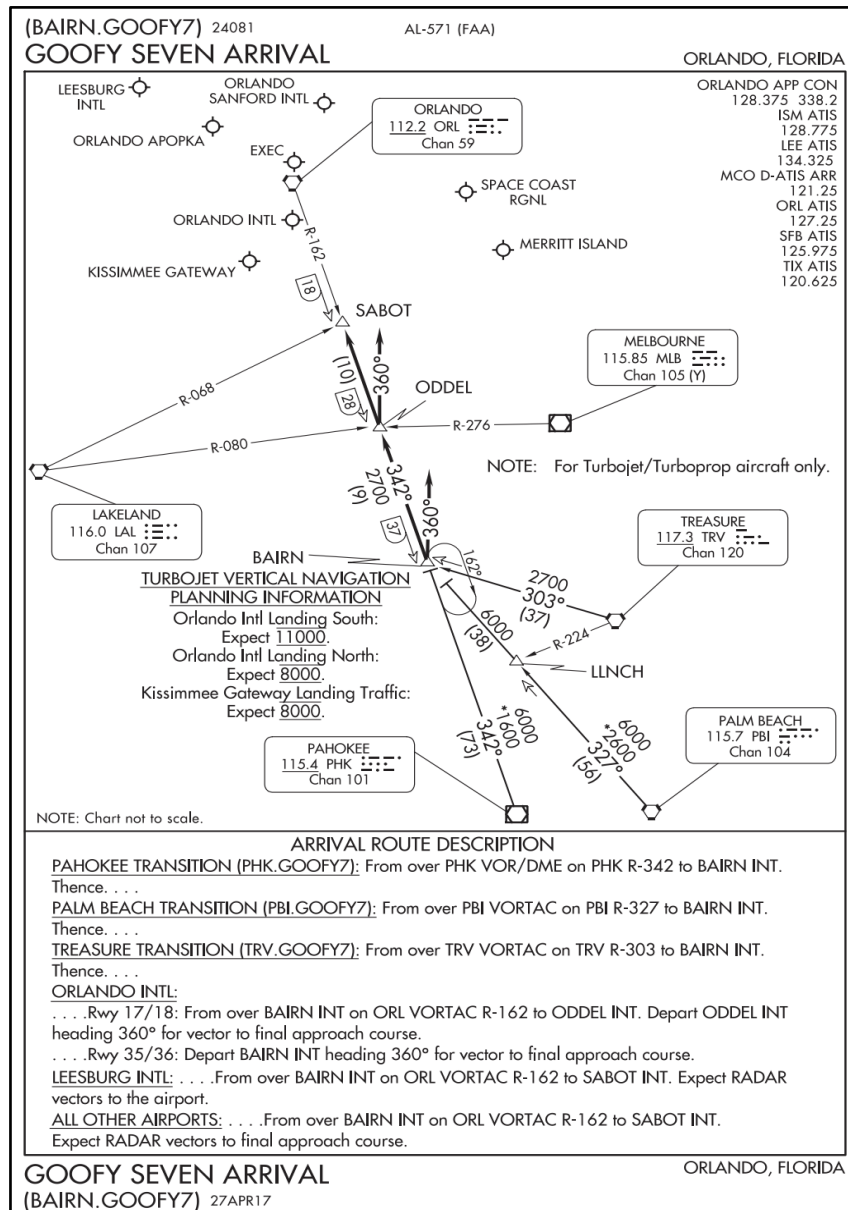
DIVERSE VECTOR AREA (RADAR VECTORS)
ORIG 13NOV14 (14317) (FAA)
Rwys 25L, 25R, heading as assigned by ATC; requires minimum climb of 370' per NM to 4600.

- Visual Climb Over Airport (VCOA)
 - As the name suggests, a VCOA is a departure option for an IFR aircraft involving a climb while remaining over the field, operating in VMC equal to or greater than the specified visibility and ceiling to altitude where the aircraft can be in IMC and proceed via radar vectors or a fix mentioned in the departure procedure.
 - A departure option for an IFR aircraft when there are obstacles 3 miles away from the airport that require a climb gradient of greater than 200 ft/min.

LAKE PLACID, NY
LAKE PLACID (LKP)
TAKEOFF MINIMUMS AND (OBSTACLE) DEPARTURE PROCEDURES
AMDT 1 10OCT19 (19283) (FAA)
TAKEOFF MINIMUMS:
Rwy 14, NA-Obstacles.
Rwy 32, std. w/min. climb of 485' per NM to 4000, or 3600-3 for VCOA.
DEPARTURE PROCEDURE:
Rwy 32, climbing left turn to heading 304° to 5000 before proceeding on course.
VCOA:
Rwy 32, obtain ATC approval for VCOA when requesting IFR clearance. Climb in visual conditions to cross Lake Placid airport at or above 5200 before proceeding on course.

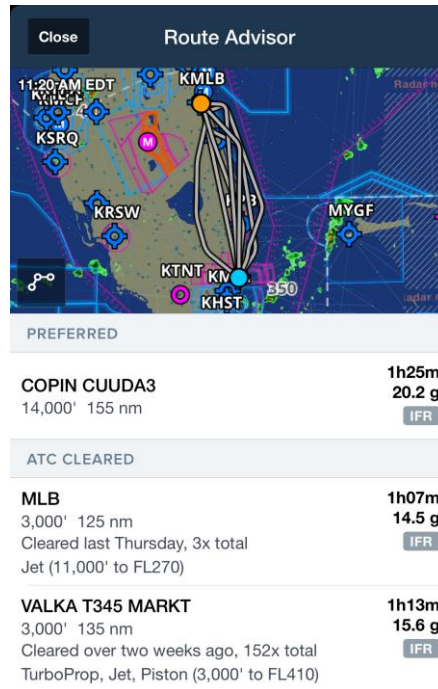
- Takeoff Obstacle Notes:
 - Low close-in obstacles are noted in the IFR Takeoff Minimums, Departure Procedures, and Diverse Vector Area (Radar Vectors) section .
 - Low close-in obstacles are those obstacles that are located within 1 NM of the DER and penetrate the 40:1 Obstacle Clearance Surface (OCS).

- Arrival procedures:
 - Standard Terminal Arrival Routes (STARs):
 - These arrival routes simplify clearance delivery procedures for arriving aircraft, specifically for airports that have a high density of traffic.
 - STARs are located in the TPP.
 - Textual and graphical.
 - A pilot may be issued a clearance containing a STAR.
 - To accept this clearance, a pilot must at least have the textual description.
 - It is the pilot's responsibility to accept or deny the issuance of a STAR.
 - A pilot may file an IFR flight plan and put "NO STAR" in the remarks.
 - If a STAR is accepted, it requires mandatory compliance.
 - Demonstrate to the student how to find and brief a STAR.



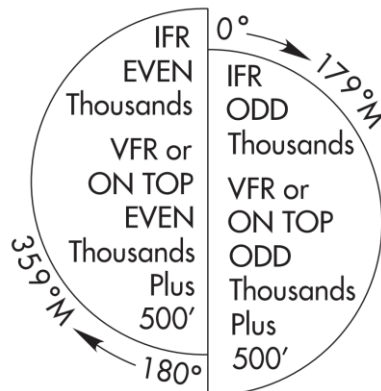
- **Route Selection:**

- ATC preferred routes
 - Located in the chart supplement
 - Check requirements of airways, SIDs, and STARs to determine if they can be flown
- EFB
 - Some flight planning apps/services can create routes as well as show how often they're cleared, the aircraft and altitudes the route is designed for, etc.



- Create your own route
 - Use airways as much as possible
 - Check requirements of airways, SIDs, and STARs to determine if they can be flown

- **Altitude Selection:**

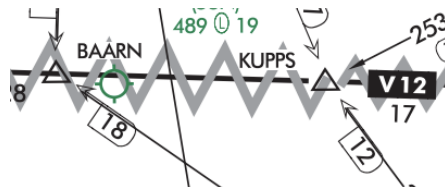


Fuel Requirements

- 91.167 - Fuel requirements for flight in IFR conditions:
 - Departure to destination airport
 - +
 - Most distant alternate
 - +
 - 45 minutes at normal cruise
- FIT FOM 3.4.7 Fuel Reserves - Usable Fuel:
 - One hour of reserve at all times

Low Altitude Enroute Chart

- IFR Enroute Low Altitude Charts provide aeronautical information for navigation under instrument flight rules below 18,000 feet MSL.
 - Updated every 56 days.
 - Airports:
 - Text is read nearly the same as the VFR sectional, with the inclusion of a “C” or “D” to indicate Class C or D airspace.
 - Types of airports:
 - Airports in **green** have instrument approach procedures.
 - Airports in **blue** have high-altitude military approach procedures.
 - Airports in **brown** do not have a published approach procedure.
 - Airspace:
 - Blue shaded region indicates Class B or C airspace.
 - Special use airspace is depicted with a blue or brown dashed outline.
 - Airways:
 - Airways are like “highways” in the sky, used to route traffic efficiently from place to place.
 - Usually 8nm wide (4nm each side).
 - Designed to provide 1,000ft of separation from IFR traffic and 500ft of separation from VFR traffic.
 - Types of airways:
 - Victor airways are based on VORs and are depicted in **Black**.
 - Tango airways are based on GPS waypoints and are depicted in **Blue**.
 - Bravo or Alpha airways are based on NDBs and are depicted in **Amber**.
 - Some airways may be joint - more than one type of navigation can be used for a segment.
 - Unusable segments of an airway will be depicted as shown:



○ IFR altitudes:

■ Minimum IFR altitude (91.177)

- Except for takeoff or landing... no person may operate an aircraft under IFR below:
- Minimum altitudes prescribed for the flown segment, or if none:
- **Mountainous areas: 2,000ft** above the highest obstacle within a horizontal distance of **4nm** from the course
- **Non-mountainous areas: 1,000ft** above the highest obstacle with a horizontal distance of **4nm** from the course

■ Minimum Enroute Altitude (MEA): assures acceptable navigational signal coverage and meets obstacle clearance requirements.

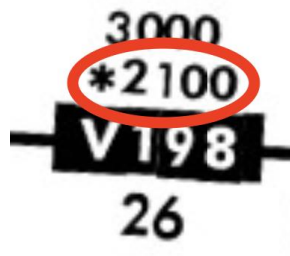
- Depicted as a number above an airway segment.



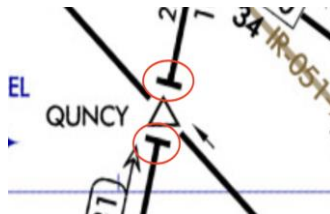
- GPS MEA: depicted in **Blue** for Tango (GPS) airways.
- MEA GAP: an area of loss in navigational coverage.

■ Minimum Obstruction Clearance Altitude (MOCA): meets obstacle clearance requirements and assures acceptable navigational signal coverage only within 22nm of a VOR.

- Depicted as a number with an asterisk * above an airway segment.



■ "Ts" on the end of airway segments passing through a fix indicate a change in the MEA, MOCA, and/or MAA.

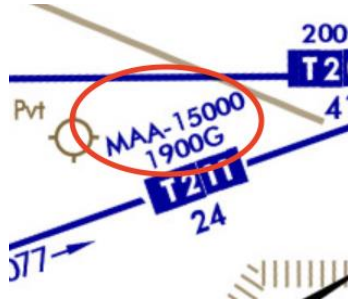


■ Off Route Obstruction Clearance Altitude (OROCA): provides obstacle clearance only.

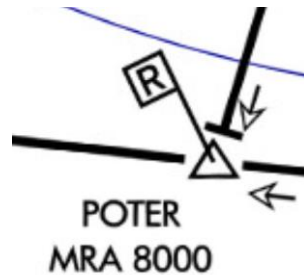
- Similar to the Maximum Elevation Figure found on VFR sectional, an OROCA is depicted for each quadrant.



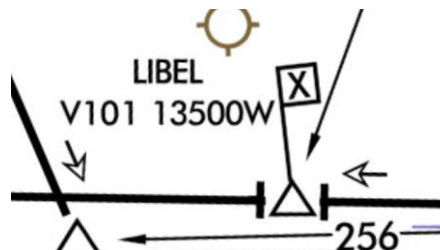
- **Maximum Authorized Altitude (MAA):** maximum usable altitude or flight level for an airspace structure or route segment.



- **Minimum Reception Altitude (MRA):** minimum altitude the navigation signal for the route and for off-course NAVAID facilities that determine a fix.

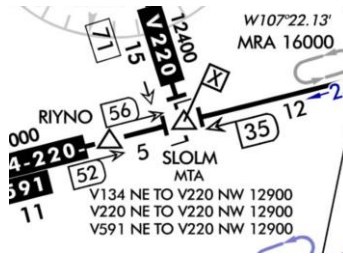


- **Minimum Crossing Altitude (MCA):** lowest altitude at certain fixes at which the aircraft must cross when proceeding in the direction of a higher minimum enroute IFR altitude.



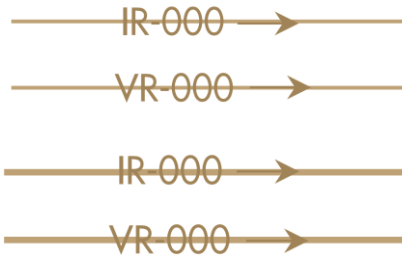
- **Minimum Turning Altitude (MTA):** lowest altitude at certain fixes where aircraft are expected to turn from one airway onto another. Guarantees obstacle protection should the aircraft fly over a fix (overshoot) during the turn.

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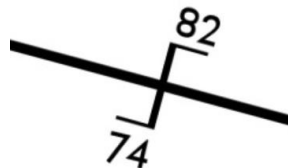


○ Miscellaneous symbology:

- Military Training Routes (MTRs): depicted as a solid brown line, indicating low-altitude, high-speed training.
 - All IR (IFR) and VR (VFR) MTRs are shown except for VRs at or below 1,500ft AGL.
 - Thin lines: 5NM or less both sides of centerline.
 - Thick lines: greater than 5NM either or both sides of centerline.



- VOR changeover point: indicates the location along an airway (depicted as DME distances along an airway) at which the aircraft must change VOR frequency.
 - If not depicted, change VOR frequency at the midpoint or bend of an airway.



- Localizer course with additional navigation function indicates a localizer which may also be used to identify a fix on an airway.
 - In this example, the I-BQK localizer may also be used to identify BERTT, a fix on T425:



- MON designation: indicates the airport has at least one instrument approach procedure not requiring the use of GPS.

MON
Miami Exec
(TMR)

Clearances

Most of this section (except for MARVELOUSVFR500) should be a review from Lesson 318. Consider quizzing the student to evaluate their knowledge.

- When is a clearance required?
 - According to 14 CFR 91.173, no person may operate an aircraft under IFR in controlled airspace unless an IFR flight plan has been filed and a clearance has been received from ATC.
 - Although flying into IMC in uncontrolled airspace without a clearance is technically legal, it has been found in the past to be a violation of **91.13 - Careless or reckless operation**.
- Obtaining an IFR clearance
 - Towered airport: Contact clearance delivery or ground frequency. The ATIS will sometimes specify a frequency.
 - Non-towered airport: Consult chart supplement / AFD for frequency or ARTCC phone number.
 - Depart VFR and contact ATC.
 - Contact ATC on the ground. Expect a *Clearance Void Time* or *EDCT (Expect Departure Clearance Time)*.
 - Clearance void time: The latest time at which one may depart IFR. A pilot who does not depart prior to the clearance void time must advise ATC as soon as possible of their intentions.
 - Ground communication outlet (GCO): on the ground, click mic 6 times to contact FSS & 4 times to contact ATC.
- Elements of an IFR clearance:
 - **C:** Clearance Limit
 - **R:** Route
 - **A:** Altitude
 - **F:** Frequency
 - **T:** Transponder

TECH FLIGHT 635, Cleared to the Sarasota International Airport as filed, after departure maintain 2000ft, expect 6000ft 10 minutes after departure, departure frequency is 132.65, squawk 0812.

- Amended IFR clearance
 - Amendments to the initial clearance will be issued at any time an air traffic controller deems such action necessary to avoid possible conflicts between aircraft.
 - Clearances will require that a flight “hold” or change altitude prior to reaching the point where standard separation from other IFR traffic would no longer exist
- Cruise clearance:
 - The term “cruise” may be used instead of “MAINTAIN” to assign a block of airspace to a pilot from the minimum IFR altitude up to and including the altitude specified in the cruise clearance.
 - The pilot may level off at any intermediate altitude within this block of airspace. Climb/descent within the block is to be made at the discretion of the pilot. However, once the pilot starts descent and verbally reports leaving an altitude in the block, the pilot may not return to that altitude.
- Approach clearance:
 - Authorization by ATC for a pilot to conduct an instrument approach. The type of instrument approach for which a clearance and other pertinent information is provided in the approach clearance when required.
 - “Fly heading 060, maintain 2000 until established, cleared for the ILS 09R.”
- Required position reports under IFR (MARVELOUSVFR500) - AIM 5-3-3, 91.183, 91.187:
 - Missed approach
 - Airspeed ± 10 kts / 5% change of filed TAS (whichever is greater)
 - Reaching a holding fix (report time & altitude)
 - VFR on top when an altitude change will be made.
 - ETA changed ± 2 min, or ± 3 min in North Atlantic (NAT) *
 - Leaving a holding fix/point
 - Outer marker (or fix used in lieu of it) *
 - Unforecasted weather
 - Safety of flight
 - Vacating an altitude / FL
 - Final Approach fix *
 - Radio/Nav/approach equipment failure
 - Compulsory reporting points ▲ *
 - 500 - unable to climb/descend at 500 fpm

* Only required in non-radar environments (acronym: COFE)

Alternate Requirements

- When is an alternate required (91.169)?
 - An alternate is NOT required when the DESTINATION airport:
 - Has an instrument approach procedure

AND

- Weather reports and forecasts indicate that from 1 hour before to 1 hour after ETA, the ceiling will be at least 2,000' AGL and the visibility will be at least 3SM (1,2,3 rule).
- Standard Alternate Minima - 91.169:
 - Precision: 600 ft ceiling and 2 statute miles.
 - Non-precision: 800 ft ceiling and 2 statute miles.
 - No instrument approach available: descent from the MEA and landing must be conducted under basic VFR.
-  designation on the approach chart indicates that the approach procedure has non-standard minimums (for aircraft other than helicopters) or restrictions (for all users) for its use as an alternate.

 ALTERNATE MINS	
24081	
NAME	ALTERNATE MINIMUMS
MELBOURNE, FL	
MELBOURNE ORLANDO	
INTL (MLB).....	ILS or LOC Rwy 9R ¹²
	LOC BC Rwy 27L ¹³
	RNAV (GPS) Rwy 9L ³
	RNAV (GPS) Rwy 9R ³
	RNAV (GPS) Rwy 27L ³
	RNAV (GPS) Rwy 27R ³
	VOR Rwy 9R ³
	VOR Rwy 27L ³
¹ NA when control tower closed.	
² LOC, Category D, 900-2¾.	
³ Category D, 900-2¾.	

-  NA designation on the approach chart means that pilots may not use that approach as an alternate due to unmonitored facility, absence of weather reporting service, or lack of adequate navigation coverage.
- Choosing an alternate:
 - Ensure that fuel requirements can be complied with (91.167)
 - Make sure that the airport has approaches that are approved to be used as an alternate (does not have the  NA).
 - Check for any non-standard alternate minima.
 - Is your aircraft equipped with WAAS? If yes, then the alternate and destination airports can be solely equipped with GPS based approaches. If not, then either the destination or alternate must be equipped with a ground-based instrument approach system (AIM 1-1-17).
 - Consider runway length, lighting, services available, etc.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.

- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 326.
 - Assign a cross-country flight plan (>50nm).
- Assign Homework:
 - Create IFR navlogs for the assigned cross-country.
 - Review NOTAMs for the assigned cross-country airports and their approaches.

Completion Standards

This lesson will be complete when:

1. Through oral quizzing and discussion, the student displays a complete understanding of weather.
2. The student has a basic understanding of IFR cross-country planning.